B Why every required refinement is optimal

Qualified half-cost argument; one fixed law, loss and feasible family

Required premise

g^* is an EXACT minimizer at $\tau = 1/2$. EVERY required child is feasible.

Weak: full unit refinements

$\bar{e}_{ij} + \bar{e}_{ji} = 1$
Full unrestricted $\mathcal{A}_n$.
All strict unit-order children.
$q = M - N(g^*)$

Neutral: full class refinements

$c_{ij} + c_{ji} = \delta_{ij}$
Full active-profile equality classes.
All class-strict children feasible.
Original unit-pair multiplicities.
$q = T - Q(g^*)$

Uniform average over ALL required children

Mean added error $= q/2$; half-cost withholding reduction $= q/2$.
Mean $R(r) = R(g^*) = R^*$

Exact optimum + feasible children

$R(r) - R^* \geq 0$ for every child.

Uniform mean of all child gaps

$$\frac{1}{L}\sum_{\ell=1}^{L}[R(r_\ell) - R^*] = 0$$

Both facts together ⇒ every required child has $R(r) = R^*$

Endpoints: weak $N = M$; neutral $Q = T$ (raw N may remain below M).
Guarded $T = 0$: one class/all-withhold, point frontier; Q/T undefined.

Uniform averaging is a proof device, not a reporting policy. Caps, fixed orders, protections and T11 need separate arguments. No arbitrary-refinement guarantee for an approximate optimizer.